Shreyas Krishnan

Research Assistant, Data Innovation & AI Lab, UC Berkeley
shreyas21447@gmail.com — shreyas.krishnan@berkeley.edu — linkedin.com/in/shreyas-krishnan-579989212

RESEARCH INTERESTS

Mechanistic interpretability and evaluation of foundation models, with a focus on data-centric and gradient-informed training for efficiency and reliability, and applications to multimodal and neuro-language settings.

EDUCATION

Birla Institute of Technology & Science (BITS) Pilani, Goa, India — Aug 2020 – Aug 2025
B.E. Electronics & Communication Engineering & M.Sc. Mathematics (Dual Degree); Minor in Data Science
Thesis Title: *Beyond modalities: robust neural representation of language in the brain.* Link

EXPERIENCE

UC Berkeley – Data Innovation and AI Lab — Berkeley, California, USA
Research Assistant : Dr. Abhishek Nagaraj — Aug 2025 – Present

- Led ORQA (first author), an occupation-indexed benchmark of 1,021 source-verified Q&A items across 183 occupations, built via an O*NET + BLS agentic pipeline; ranked 15 frontier and open-weight models, validated against GDPval and GDPval-AA Elo. Under review at NeurIPS 2026. preprint.
- Built an SAE and linear-probe interpretability-and-steering pipeline (co-first author) for LLM social-simulation agents on Llama-3.3-70B-Instruct; dense λ sweeps over probe directions (82% layer-48 accuracy) shifted lottery switching points from $\sim$30 to >180 tokens with monotone, dose-dependent control. Accepted at NBER; under review at COLM. preprint.
- Conducting research on how access to pirated books (Books3 dataset) shapes large language model performance, using a novel "name cloze" evaluation across 12,000+ books and causal identification based on publication-year variation. NBER Working Paper 33598.
- Embedded $\sim$20k books from $\sim$200 publishers (Keepa) to map semantic market positioning. Novel Positioning Measures beat genre dummies (e.g., $\sim$34% review variance), revealing Big Five cross-genre hubs and challenger advantages.

Harvard University – Kreiman Lab — Boston, Massachusetts, USA
Masters Thesis: Dr. Gabriel Kreiman — Jan 2024 – Aug 2025

- Analyzed intracranial iEEG during language tasks, trained machine learning decoders for grammatical and semantic features, and mapped region-specific effects.
- Co-designed an LLM-based real-time memory retrieval system, evaluated Gemini 2.5, GPT-o3, Leta, and Zep, and built a benchmarking and evaluation pipeline.
- Manuscripts include a Nature submission (final edits) and a NeurIPS 2025 submission (preprint linked below).

Nanyang Technological University (NTU) — Singapore
Research Intern: Dr. Amalin Prince, Dr. Yuvaraj Rajamanickam — Oct 2022 – May 2024

- Developed EEG emotion recognition with ResNet and hybrid models, achieving mean accuracy of 99.34% on DREAMER and 92.18% on DEAP, and wrote a first-author IET book chapter.
- Developed a Vision Transformer for remote hand-raise detection that achieved $93.82\% \pm 2.16\%$ (5-fold CV) and was accepted to IEEE CIACON 2025.

National Institute of Advanced Studies (NIAS) — Bangalore, India
Research Intern: Dr. Snehanshu Saha — May 2023 – May 2024

- Developed Granger-causality-guided pruning for interpretability and efficiency, resulting in a first-author IJSCAI 2025 paper.

Indian Institute of Technology Madras — Chennai, India
Research Intern: Dr. Ganapathy Krishnamurthy — Jun 2023 – Aug 2023

- Implemented FAIR-style architectures for object detection and instance segmentation, and ran ablations and analyses.

OnFinance AI — Bangalore, India
AI Engineer Intern (LLM/RAG) — May 2024 – Aug 2024

- Fine-tuned LLaMA 3 and vision-language models for financial analysis; built OCR-based extraction and automated report generation with RAG; and improved citation quality.

PUBLICATIONS

Preprints / Under Review

- D. Mayo, C. Zhang, **S. Krishnan**, A. Shaw, B. Katz, A. Barbu, B. Cheung (supervision). *Look But Don't Touch: Gradient-Informed Selection Training.* Under Submission: ICLR 2026. preprint
- **S. Krishnan**, S. Chang, A. Nagaraj. *ORQA: An Occupation Realistic Question and Answer Benchmark for LLM Professional Assistance.* Under review at NeurIPS 2026. preprint

- J. G. Fan, A. Murugan, **S. Krishnan**, A. Nagaraj. *Beyond Prompting: Using Sparse Autoencoders and Probes to Understand and Steer LLM Agents in Social Scientific Simulations.* Accepted at NBER; under review at COLM. preprint
- **S. Krishnan**, A.Thamma. *From Confusions to Confidence:Calibrating Vision Confidence With Human Confusion Priors.* Accepted at ICCV - HiCV Workshop 2025.
- **S. Krishnan**, A. Thamma (2025). *Plan–Check–Revise: A Token-Paritized Two-Agent Protocol for Verifiable Math Reasoning.* Submitted to NeurIPS 2025 Workshop MATHAI.
- S. Madan, **S. Krishnan**, G. Kreiman (2025). *Augmenting Human Memory with AI: Benchmarking Long-term Personal Memory Recall in LLMs.* Under Submission : NeurIPS 2025. preprint
- P. Misra, **S. Krishnan**, G. Kreiman (2025). *Beyond modalities: robust neural representation of language in the brain.* Under submission at Nature.
- Served as a peer reviewer for the NeurIPS 2025 MATH-AI Workshop (link) and reviewed two submissions.

Journal & Conference

- **S. Krishnan**, L.Murugan, A. Hassouneh,Y. Rajamanickam, A. Prince, T. Thiyagasundaram, M. Murugappan (2024). *Emotion Recognition using ResNet Feature Extraction on EEG Signals.* IET (UK), Book Chapter. DOI — print
- **S. Krishnan** (2025). *Vision Transformer for Hand-Raise Recognition in Remote Learning* (93.82% $\pm$ 2.16%, 5-fold CV). IEEE CIACON 2025. DOI — print
- **S. Krishnan**, A. Das (2025). *Harnessing Chaos and Causality in Neural Networks: A Pruning Strategy for Enhanced Performance and Explainability.* IJSCAI 2025. print

SELECTED COURSES

Advanced/Graduate-level Topics

- Machine Learning (Course Rep, BITS-F464, Aug–Dec 2023)
- Optimization
- Statistical Inference & Applied Statistical Methods
- Foundations of Data Science

Undergraduate Core

- Linear Algebra; Probability & Statistics; Numerical Analysis
- Digital Signal Processing (DSP)
- Multivariate Calculus; Ordinary Differential Equations
- Discrete Mathematics; Control Systems

AWARDS & DISTINCTIONS

Center for Brains, Minds and Machines, Massachusetts Institute of Technology Woods Hole, MA
Selected (fully funded) among ∼30 global graduate students/postdocs worldwide for a deep-dive course on the science of intelligence. Aug 2025

OTHER EXPERIENCES

Course Representative, Machine Learning (BITS-F464) Aug 2023 – Dec 2023

- Coordinated lectures and labs, and liaised on assignments and grading logistics.

Student Clubs & OSS 2022 – 2024

- Society of AI and Deep Learning (SAiDL): Implemented mixup on the TREC dataset (GitHub).
- Electronics and Robotics Club: Built ROS path planning using Dijkstra's algorithm (GitHub).

SKILLS

- **Programming:** Python, MATLAB, C/C++, JavaScript
- **ML/Tools:** PyTorch, TensorFlow, scikit-learn, OpenCV, Pandas/NumPy, Linux, Git, Weights & Biases, basic distributed training
- **NLP/Systems:** OCR pipelines, RAG, LLM fine-tuning and evaluation; iEEG and EEG processing
- **Design:** Adobe Illustrator, Inkscape, Photoshop

REFERENCES

Available upon request.
Dr. Gabriel Kreiman : gabriel.kreiman@tch.harvard.edu,Professor Harvard University
Dr. Hanspeter Pfister : pfister-admin@seas.harvard.edu ,Professor Harvard University
Dr.Abhishek Nagaraj : nagaraj@berkeley.edu,Associate Professor UC Berkeley